

had 5500 employees in 2022-23. By the end of 2024, they were projected to have 8000. That is 2500 new jobs even before their main campus became operational. When we add Tata, Micron, Rockwell, and others, including supply chain and support, the figure of 25,000 is quite realistic. The ripple effect is significant.

There are speculations that the government will fund universities to set up fabs. Is that the right way to build skill capacity?

It is an admirable aspiration, but we must be pragmatic. Not every institute can or should set up a semiconductor lab. I heard a director claim they need ₹2 billion for a lab—this is simply unrealistic. We already have excellent infrastructure at IISc Bangalore and select IITs. **Maybe the government will invest in a few more centres strategically, but the idea should be to insist on a hub-and-spoke model. These centres should support surrounding institutions, allowing students to access state-of-the-art facilities without unnecessarily duplicating infrastructure.**

Even with ₹2 billion, would an institute be able to replicate what a real fab does?

No. Even then, it would not match the capability of a commercial fab. Take ATMP dicing tools, for example. An entry-level diamond saw dicing machine (second-hand) might cost under \$250,000, but it will have limitations in scribe line width. Advanced fabs utilise laser-based dicing equipment, which can cost between \$1.5 million and \$2 million. And such tools become obsolete in just a few years. **You cannot chase fab-grade specs. What you can do is expose students to the process: defect identification, dicing parameters, slurry composition, and yield factors.** That is more practical than running an actual fab in the Institute to train people.

So, rather than every institute investing in heavy capital expenditure, they should focus on partnering with hub institutes?

Absolutely. Let a few nodal institutions build and maintain the facilities. The operational expenditure for cleanrooms and specialised tools is very high. It is impractical for smaller colleges to manage this for just a few student batches. Instead, we should encourage the use of shared infrastructure, regional workshops, and inter-institutional access.

Should the government require institutes applying for grants to show tie-ups with other colleges and clear student targets?

Yes, and that is already happening. Under programmes such as CHIP to START (C2S) or INUP, MeitY mandates that grantees must train additional people and have a commercial or scalable outlook. This requirement is built into the mandate. The government is committed to broadening the talent base, rather than letting capability remain siloed within a single campus.

But how will these institutes manage high running costs while keeping access affordable?

They return to the government for operational expenditure support. That is part of the ongoing conversation. Institutions such as IITs and IISc are not profit-driven; they are centres of excellence and fountains of knowledge. The government is willing to fund them because it trusts their intent and recognises the national importance of their mission.

Is this assistance only for government or autonomous colleges? What about private colleges?

The government has been quite generous. Many private colleges now have access to EDA tools and related support. If they align with

the prescribed curriculum and engage with the broader ecosystem, through labs, site visits, or collaborative training, they can deliver excellent education.

For instance, taking students from Manav Rachna to a smart ID card facility, EMS operations, or companies like Sahasra or CDIL, and having them observe for a few days is far more impactful than months spent in a classroom.

Is there room for private companies or startups to enter this training space, not as CSR, but as viable ventures?

Some are already exploring this. In the design training space, it is happening. In manufacturing, a few companies have shown interest, although I am unsure how cost-effective their models are. That said, we have seen this approach work for design—many institutes outsource training to third-party providers until their faculty are ready to teach independently. The same could happen in manufacturing, though we must be cautious of high fees without matching infrastructure.

Any final advice or key message for institutes trying to support the semiconductor mission?

Yes. It is all hands on deck. **Institutes must accept that change is here and respond with urgency. You do not need to invest ₹100-200 million in equipment. Leverage existing hubs. Focus on student exposure. Do not let budget or infrastructure become an excuse.** We had a foreign university offer a course at ₹1 million to 1.5 million per student. Even a scaled-down version was ₹600,000-700,000. ESSCI's semiconductor skills committee concluded that we can achieve better outcomes at far lower costs by utilising our IITs and IISc. The infrastructure is already in place. Let us use it wisely, for the benefit of our students. **EFY**